E-commerce Customers, Orders & Products

STUDENT PROJECT BRIEF • Python Data Analysis • Suggested duration: 2 hours

1. Business Scenario

An online store has separate customer, order, and order-item tables. Management wants a joined view of customer segments, order behavior, product categories, and profitability.

2. Project Mission

Combine related tables correctly, analyze customer and product performance, and build a dashboard without double-counting order-level values.

3. Learning Objectives

- Inspect a real-world-style dataset and explain its structure and limitations.
- Use NumPy and Pandas for data types, missing values, duplicates, derived columns, grouping, and aggregation.
- Use Matplotlib, Seaborn, and/or Plotly to communicate patterns with appropriate charts.
- Translate findings into practical, evidence-based business recommendations.
- Build a simple interactive Streamlit dashboard for a non-technical stakeholder.

4. Provided Data

customers_raw.csv — Customer-level table: customer ID, city, customer segment, and signup date.

orders_raw.csv — Order-level table: order ID, customer ID, order/delivery dates, sales channel, payment method, order status, and shipping fee.

order_items_raw.csv — Item-level table: order ID, category, product, quantity, price, discount, cost, line sales, line cost, and line profit.

5. Required Tasks

- **Task A — Understand the data:** load the file(s), inspect shape, columns, data types, sample rows, and summary statistics.
- **Task B — Clean the data:** Inspect each table separately for missing values, duplicates, invalid IDs, date issues, and numeric anomalies. Validate key uniqueness and referential integrity before joining.
- **Task C — Explore and analyze:** answer the business questions below using suitable groupby/aggregation and at least one time-based analysis where dates are available.
- **Task D — Visualize:** create at least 4 meaningful charts. Add clear titles, labels, units, and readable formatting.
- **Task E — Recommend:** write 3–5 prioritized recommendations. For each, cite the finding that supports it and mention any limitation.
- **Task F — Build the dashboard:** use Streamlit to present KPIs, charts, and at least two interactive filters where appropriate.

6. Business Questions to Answer

- How do order volume and sales vary by month, channel, city, and customer segment?
- Which product categories and products contribute most to sales and profit?
- How do order status and delivery timing vary across segments or channels?
- Which customer segments show stronger order value or repeat-order behavior, if it can be derived reliably?
- What customer or product actions should the business test next?

7. KPI Definitions / Analysis Focus

Order count; customer count; line sales; line profit; profit margin; average order value; delivery days; status mix.

8. Suggested Visualizations

Monthly orders/sales; sales and profit by category; customer-segment comparison; channel/payment comparison; delivery/status analysis.

9. Important Project-Specific Rule

Critical join rule: aggregate item-level data by Order_ID before merging item totals into the order-level table. Joining raw item rows directly to orders duplicates order-level values such as Shipping_Fee_EGP. State the grain of every table and metric.

10. Recommended Workflow (120 minutes)

Time	Activity	Expected output
0–10 min	Understand scenario and inspect dataset(s)	Data dictionary / initial observations
10–30 min	Clean data and document decisions	Cleaned dataframe(s)
30–60 min	Calculate KPIs and explore patterns	Analysis tables and findings
60–85 min	Build and refine visualizations	At least 4 charts
85–110 min	Build Streamlit dashboard	Working dashboard with filters
110–120 min	Summarize recommendations and prepare demo	Short presentation

11. Deliverables

- A Jupyter Notebook (.ipynb) or Python analysis script with readable, commented code.
- A working Streamlit app (app.py) with KPIs, visualizations, and filters.
- A cleaned dataset or a reproducible cleaning pipeline; do not overwrite the original raw data.
- A one-page executive summary containing 3–5 findings, 3 recommendations, and limitations.
- A short 3–5 minute presentation/demo explaining the business problem, analysis, dashboard, and recommendations.

12. Submission and Technical Requirements

- Use Python, Pandas, NumPy, and at least one of Matplotlib, Seaborn, or Plotly.
- Use Streamlit for the dashboard. Include a requirements.txt file and clear run instructions.
- Make the project reproducible: use relative paths, descriptive variable names, and avoid hard-coded machine-specific paths.
- Never claim correlation proves causation. Explain assumptions and limitations.
- Submit all code and supporting files in one clearly named folder or ZIP.

13. Assessment Rubric

Criterion	Weight
Table understanding, joins, and cleaning	25%
Analysis and business metrics	25%
Visual storytelling and chart choice	15%

Criterion	Weight
Streamlit dashboard usability	20%
Recommendations and presentation	15%
Total	100%

14. Viva / Discussion Questions

- What was the most important data-quality issue you found, and how did you handle it?
- Which KPI best answers the business problem, and why?
- Why did you choose each chart, and what could make it misleading?
- Which finding is strongest, and what additional data would validate it?
- If you had one more day, what would you improve in the analysis or dashboard?

15. Final Submission Checklist

- I checked the shape, data types, missing values, and duplicates.
- I documented cleaning decisions and kept raw data unchanged.
- I validated KPI formulas and checked for division-by-zero or double-counting issues.
- My charts have clear titles, labels, units, and readable scales.
- My dashboard runs from the provided instructions.
- My recommendations are linked to findings and acknowledge limitations.